



Predicting Diabetes Risk with Explainable AI

An end-to-end machine-learning project on 253,680 US adults

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THE PROBLEM

Can we flag diabetes risk early from routine health data?

- Goal: predict whether an adult has diabetes from everyday health and lifestyle indicators (blood pressure, BMI, general health, age...).
- Data: CDC BRFSS 2015 - a survey of 253,680 US adults, 21 features.
- The catch: only ~14% have diabetes. On imbalanced data, 'accuracy' is misleading - a model guessing 'no' for everyone scores 86%.
- So the real challenge is catching true cases without crying wolf.



APPROACH

An honest, end-to-end ML workflow

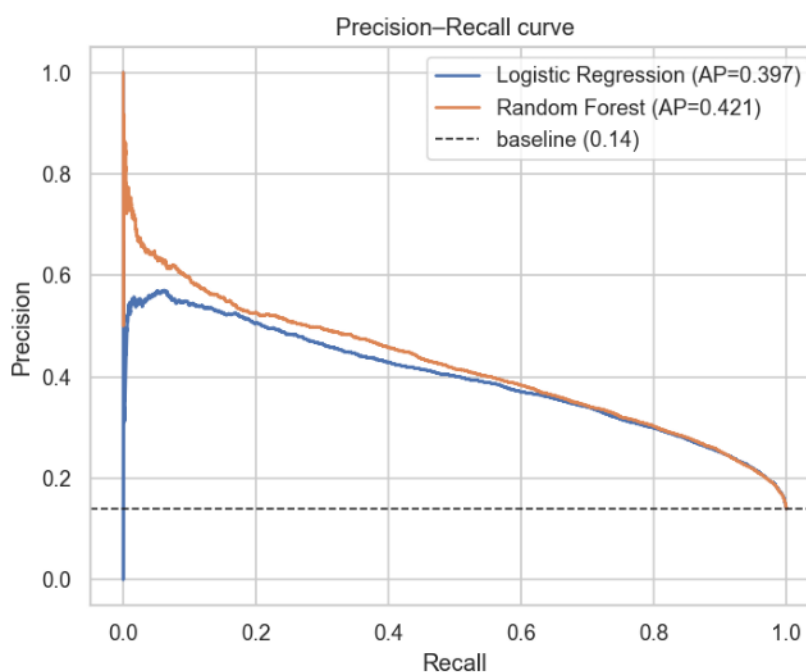
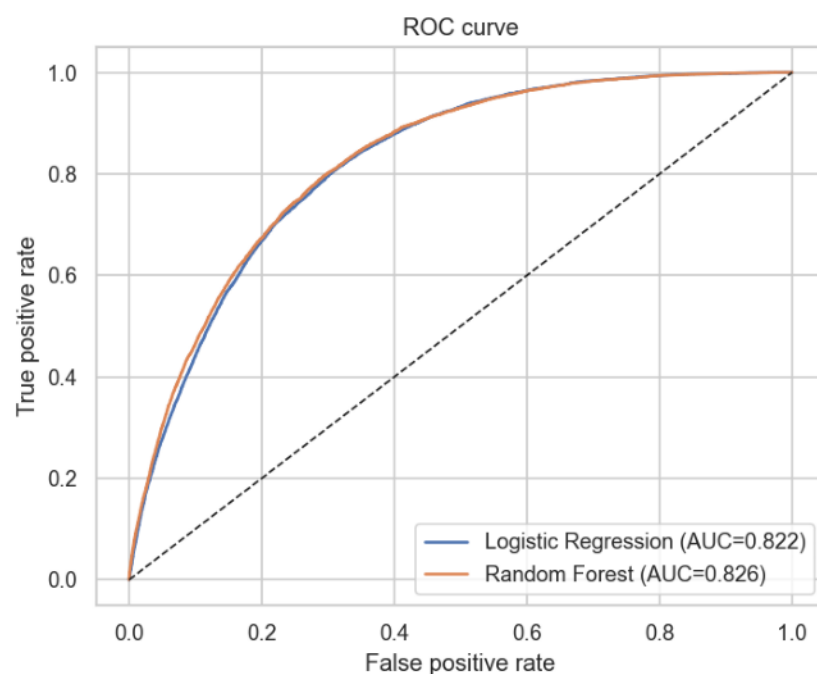
- Exploratory analysis - distributions and correlations with the target.
- Models - logistic regression (baseline) and a random forest.
- Imbalance-aware evaluation - PR-AUC, calibration, and a recall-tuned decision threshold instead of plain accuracy.
- Explainability - SHAP, to show what drives each prediction.
- Responsible framing - a screening aid, not a diagnostic tool.

RESULTS

Strong discrimination on imbalanced data

Random Forest: ROC-AUC 0.83 · PR-AUC 0.42

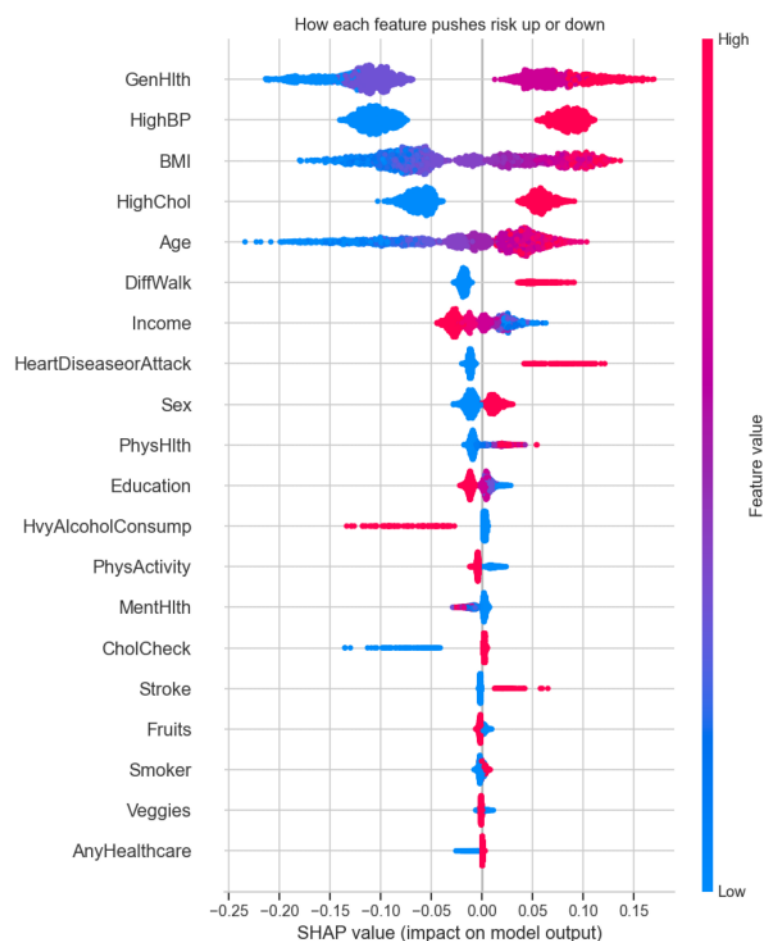
(PR-AUC vs. a 0.14 random baseline; ~62% of true cases flagged at the chosen threshold)



EXPLAINABILITY (SHAP)

Why the model predicts risk

Top drivers - general health, high blood pressure, BMI, cholesterol, and age - match the clinical literature: the model learned real signal.





TAKEAWAYS

What this project shows

- End-to-end ML: from raw survey data to an explained, evaluated model.
- Rigor on imbalanced health data - the metrics that actually matter.
- Explainable AI that a clinician could interrogate, not a black box.
- Built responsibly, with clear limits on how it should be used.

Full code, figures & write-up:

github.com/Kingsley-amg/diabetes-risk-prediction